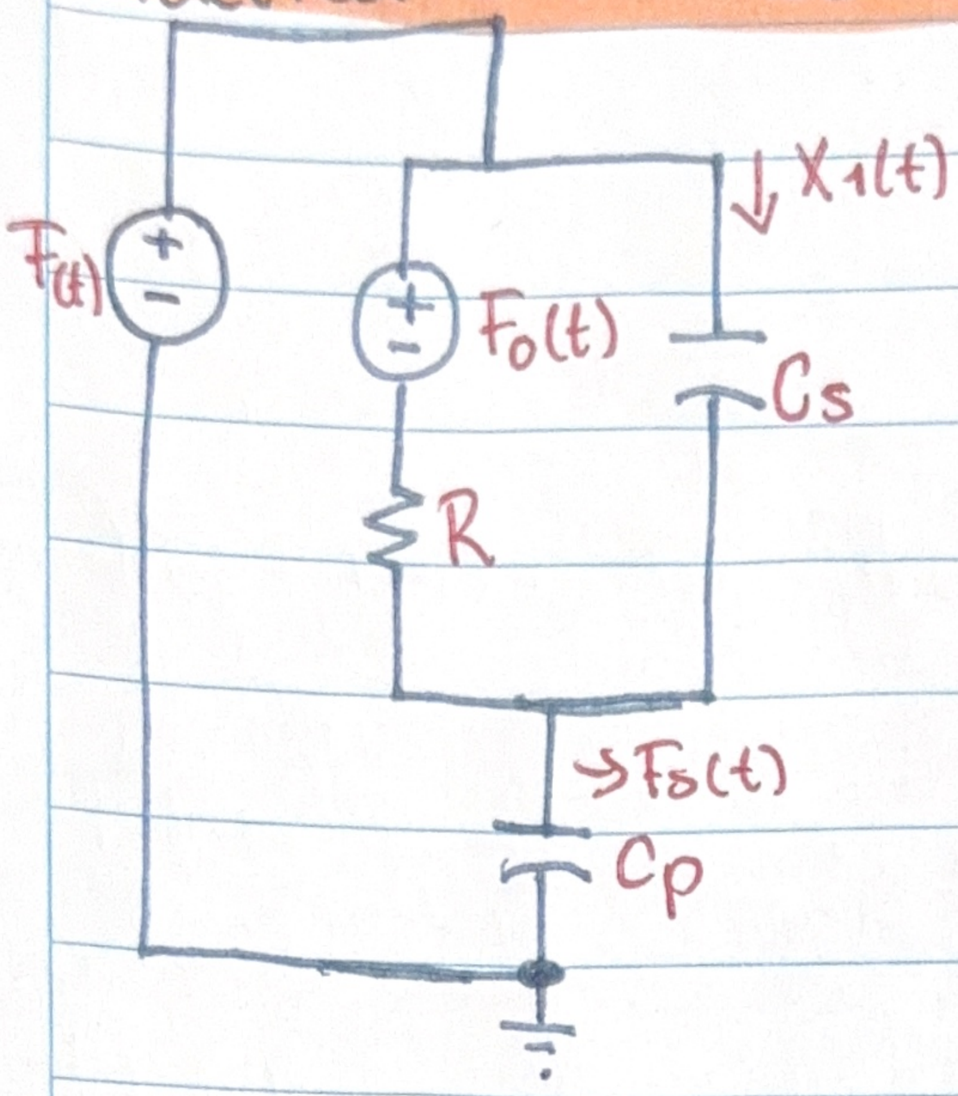
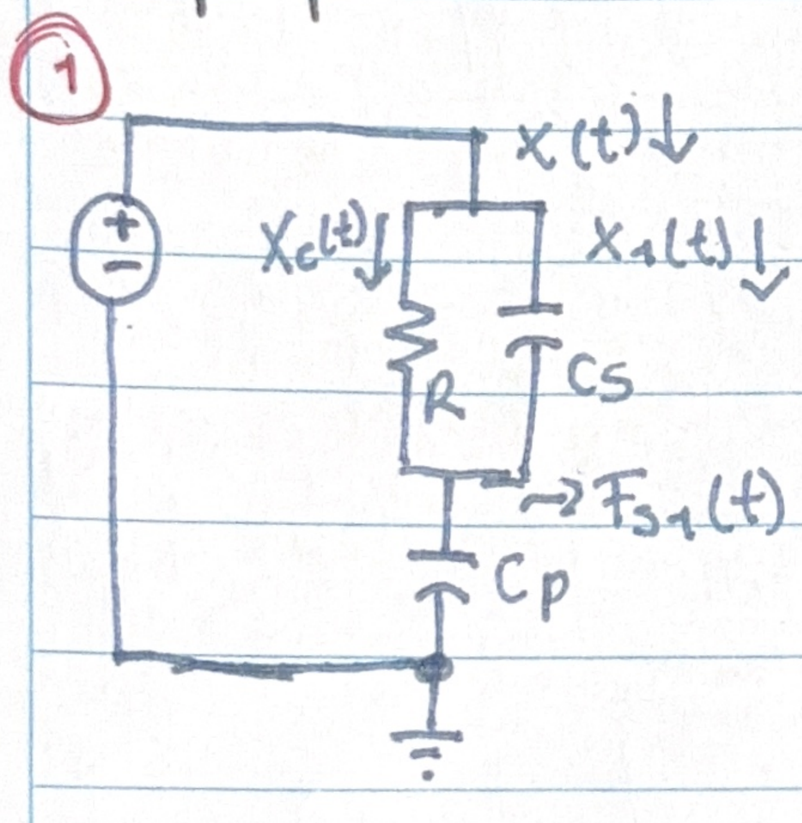
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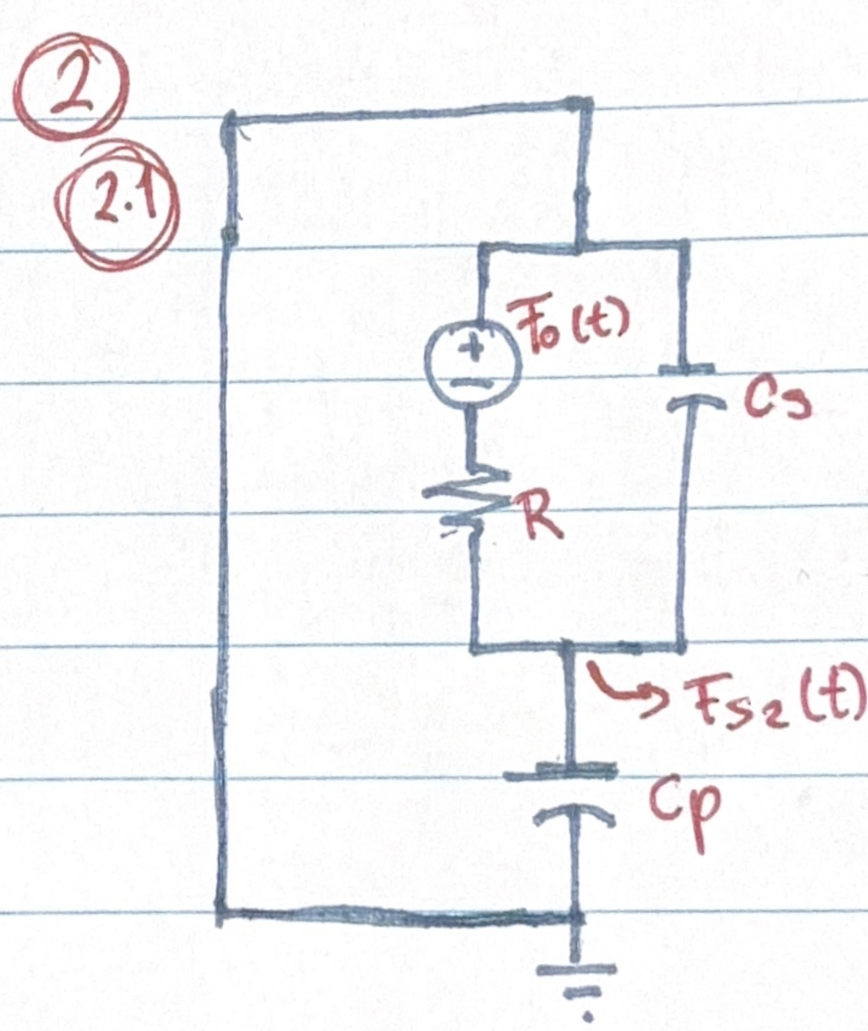
28/Oct/26

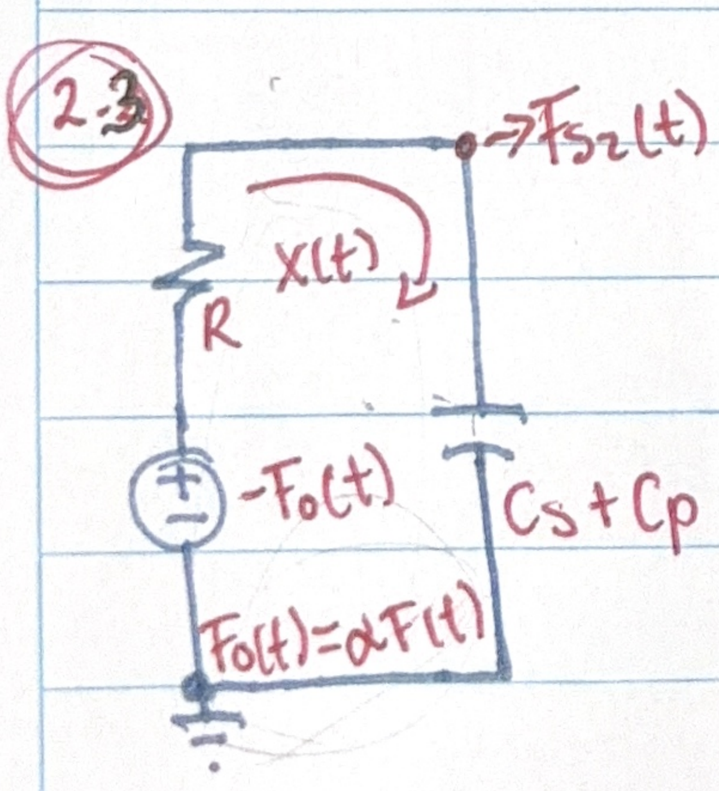
Práctica 5.5.- Modelado y control del sistema musculoesquelético.

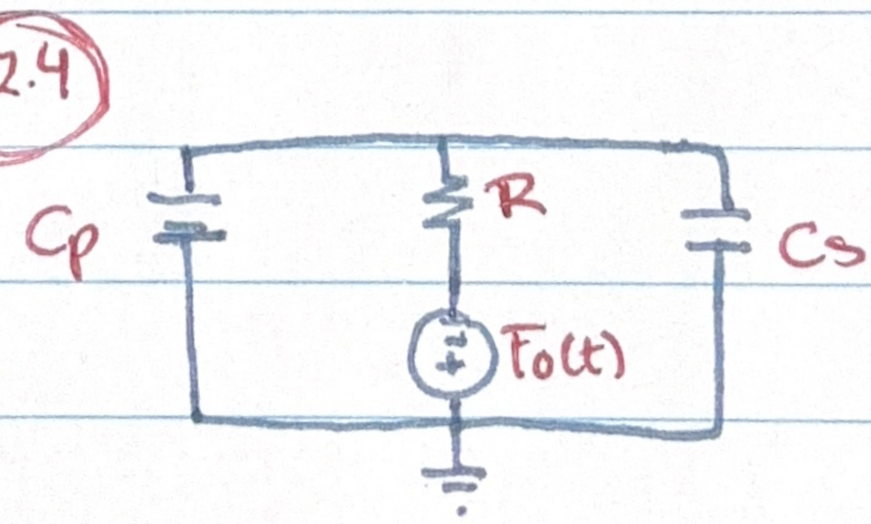


Superposición









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Analisis del circuito por superposición

* Circuito uno: apagando $F_0(t)$.

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_{ps} F_{s1}(S) = C_s S [F(S) - F_{s1}(S)] + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_{ps} F_{s1}(S) = C_s S F(S) - C_s S F_{s1}(S) + \frac{F(S)}{R} - \frac{F_{s1}(S)}{R}$$

$$\left(C_p S + C_s S + \frac{1}{R}\right) F_{s1}(S) = \left(C_s S + \frac{1}{R}\right) F(S)$$

$$\frac{R(C_p + C_s)S + 1}{R} F_{s1}(S) = \frac{C_s R_s + 1}{R} F(S)$$

$$F_{s1}(S) = \frac{\frac{C_s R_s + 1}{R} F(S)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(S)$$

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*Circuito 2: Apagando F(t)

$$-\alpha F(t) = Rx(t) + \frac{1}{Cp+Cs}\int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{Cp+Cs}\int x(t)\,dt$$

$$-\alpha F(s) = Rx(s) + \frac{X(s)}{(Cp+Cs)s}$$

$$F_{s2}(s) = \frac{X(s)}{(Cp+Cs)s}$$

$$F_s = -\frac{X(s)}{\alpha}\,\frac{R(Cp+Cs)+1}{(Cp+Cs)s}$$

$$\frac{F_{s2}}{F_s} = \frac{\frac{X(s)}{(Cp+Cs)s}}{-\frac{X(s)}{\alpha}\,\frac{R(Cp+Cs)+1}{(Cp+Cs)s}} = -\frac{\alpha}{R(Cp+Cs)s+1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R(Cp+Cs)s+1}$$

Función de transferencia

$$F_{s(s)} = F_{s1}(s) + F_{s2}(s) = \frac{CsRs+1}{R(Cp+Cs)s+1}F(s) - \frac{\alpha F(s)}{R(Cp+Cs)s+1}$$

$$F_s(s) = \frac{CsRs+1-\alpha}{R(Cp+Cs)s+1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{CsRs+1-\alpha}{R(Cp+Cs)s+1}$$